GROUP 2

Crop Recommendation System

Track	Machine Learning
Duration	2 Weeks (10 working days, 4 hours/day)
ML Task	Classification
Target	crop / recommended crop name
User Interface	Required

Project Analysis

Farmers may select crops mainly from experience even when soil and environmental conditions differ. This project requires students to build a machine-learning system that recommends a suitable crop from soil and environmental inputs.

The application should accept soil nutrient and climate values, run them through the trained model, and clearly display the recommended crop.

What the Student Must Build (Mandatory)

- Load and inspect the crop dataset.
- Check missing values, duplicates, data types, and target distribution.
- Perform EDA on soil and environmental variables.
- Use N, P, K, temperature, humidity, pH, and rainfall as core inputs where available.
- Prepare features and target.
- Split data into training and testing sets.
- Train at least two suitable classification models.
- Evaluate and compare model performance.
- Select and save the best model.
- Build a UI where the user enters soil/environment values.
- Display the recommended crop clearly.

Technical Requirements

- Python
- Pandas
- NumPy
- Scikit-learn
- Joblib/Pickle
- Git & GitHub
- Streamlit/Gradio or another suitable UI framework

Dataset Requirements

- Use the instructor-approved Kaggle crop recommendation dataset.

- Target: crop / recommended crop name.
- Core inputs: N, P, K, temperature, humidity, pH, rainfall.
- Students should document the exact crop labels present in the provided dataset instead of assuming labels not in the file.

Optional Features (Bonus)

These features may improve the project but are not required for basic project completion.

- Soil analysis charts
- Fertilizer recommendation
- Weather integration
- Crop information page

References and Starting Resources

- Kaggle - Crop Recommendation Dataset
- FAO Open Data
- Scikit-learn Documentation